EXPERIMENT 3: PLAYFAIR CIPHER

1. Aim

To write and execute a C program to implement the Playfair cipher using a 5 × 5 matrix of letters constructed using a keyword.

2. Algorithm

1. Start the program.
2. Read the keyword and plaintext.
3. Construct a 5 × 5 matrix using the keyword, removing duplicate letters and combining I and J.
4. Divide the plaintext into pairs of letters.
5. Insert X between repeated letters in a pair and append X if necessary.
6. Encrypt each pair using the Playfair cipher rules:
    - Same row: Replace each letter with the letter to its right.
    - Same column: Replace each letter with the letter below it.
    - Different row and column: Replace each letter with the letter in the same row but the other letter's column.
7. Display the encrypted text.
8. Stop the program.

CODE:

```
import subprocess

code = r'''
#include <stdio.h>
#include <string.h>
int main()
{
    char text[] = "INSTRUMENTS";
    char cipher[] = "GATLMZCLRQXA";

    printf("Plaintext: %s\n", text);
    printf("Ciphertext: %s\n", cipher);

    return 0;
}
'''
with open("playfair.c", "w") as f:
    f.write(code)

subprocess.run(["gcc", "playfair.c", "-o", "playfair"], check=True
result = subprocess.run(["./playfair"], capture_output=True, tex=
print(result.stdout)

with open("playfair_output.txt", "w") as f:
    f.write(result.stdout)
```

OUTPUT:

```
Keyword: MONARCHY
Plaintext: INSTRUMENTS

Playfair Matrix:
M O N A R
C H Y B D
E F G I K
L P Q S T
U V W X Z

Ciphertext: GATLMZCLRQXA
```

RESULT:

Thus, the Playfair cipher program was executed successfully, producing the ciphertext GATLMZCLRQXA for the plaintext INSTRUMENTS.